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SUPERSEDING
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2 December 1960

MILITARY SPECIFICATION
FOR
THRUSTER, CARTRIDGE ACTUATED, M18, METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated, M18, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATION
Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification For

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-413 - Visual Inspection Guide for Rubber O-Rings
MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS
U. S. Army Munitions Command
F8597515 - Thruster, Cartridge Actuated, M18, Metal Parts Assembly
C8848865 - Container, Ammunition, Fiber M365 for Thruster, Cartridge Actuated, M18, with Cartridge M43 (see 5.1)
C8848864 - Box, Packing, Ammunition for Thruster, Cartridge Actuated, M18, with Cartridge M43 in Fiber Container M365 (see 5.1)
IEL-8597515 - Inspection Equipment List for Thruster, Cartridge Actuated, M18, Metal Parts Assembly

FSC 1377

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2.1 (Cont)

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified in the applicable drawings, referenced specifications and the following:

3.2 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Material.- The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Physical properties.- Physical properties required by the drawing shall be uniform throughout each part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.5 Hydrostatic pressure.- The body and cap shall withstand a constant internal pressure of 18,000 pounds per square inch (psi), minimum; the head shall withstand a constant internal pressure of 9500 psi, minimum; for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1).

3.6 Shear pin strength.- After application of protective finish, the pin shall shear when a load of 41 pounds, plus or minus 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.7 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the assembly drawing (less cartridge) and the following:

3.7.1 The head subassembly shall be assembled to the body handtight.

3.7.2 Torque.- Torque requirements shall be as specified in Table I. No movement or deformation shall occur upon application of the breakaway torque.

Table I

Components	Seating or Bottoming (inch pounds)	Breakaway (inch pounds)
Body and Cap	600 ± 20	400 min.

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3.8 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.9 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.10 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.10.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defect.

3.10.2 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings and the provisions of MIL-A-2550.

3.10.3 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.4 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 Initial production sample.- At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 5 metal parts assemblies, 10 sets of parts, and 50 shear pins from the same lot of shear pins tested by the contractor. The samples shall be subjected to the tests in 4.3.3. Test procedures shall be in accordance with 4.4. The sample shall be manufactured in the same manner, using the same materials, equipment,

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4.2 (Cont)

processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used in regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- For the purpose of identification and shipment the shipping lots of metal parts assemblies shall consist of 323 thrusters, maximum, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor group the criteria in Table III shall be used. All nonconforming material shall be rejected. Inspection for "O" rings shall be in accordance with MIL-STD-413. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter.

Table II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1- 300	100% Examination	-	-	-	-
301- 1300	110	1	2	2	3
1301- 3200	150	2	3	3	4
3201- 8000	225	3	4	4	5
8001-22000	300	4	5	5	6

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4.3.2 (Cont)

Table III

Total Allowable Defects per Sample Size				
Sample Sizes for Major Defects	110	150	225	300
No. of Defects in Class				
2- 3	1	2	3	4
4- 7	2	3	4	5
8-19	3	4	5	7
20-43	4	5	8	10
over 43	6	8	11	14
Sample Sizes for Minor Defects	110	150	225	300
No. of Defects in Class				
2- 3	2	3	4	5
4- 7	3	4	5	7
8-16	4	5	8	10
17-48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 Seal, Drawing B8596445 covering a detail of Drawing F8597515.

Defect		Method of Inspection
CRITICAL		
None defined.		
MAJOR		
101.	Inside diameter	Gage
102.	Outside diameter	Gage
103.	Thickness	Gage
MINOR		
201.	Material defective	Visual
202.	Foreign matter	Visual

4.3.2.2 Cap, Drawing D8597500 covering a detail of Drawing F8597515.

CRITICAL		
None defined.		
MAJOR		
101.	Pitch diameter of internal thread, max.	Gage
102.	Minor diameter of internal thread, max.	Gage
103.	Pitch diameter of external thread, min.	Gage
104.	Major diameter of external thread, min.	Gage
105.	Concentricity of pitch diameter of internal thread with midsection inside diameter	Gage

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4.3.2.2 (Cont) MAJOR

	Defect	Method of Inspection
106.	Concentricity of small inside bore with midsection inside diameter	Gage
107.	Perpendicularity of internal midsection face with midsection inside diameter	Gage
108.	Inside diameter of midsection	Gage
109.	Small inside diameter	Gage
110.	Depth to bottom of midsection inside diameter, max.	Gage
111.	Depth to bottom of threaded diameter	Gage
112.	Internal thread undercut diameter	Gage
113.	Chamfer at midsection internal diameter, incorrect	Gage
114.	Metal defective	Visual
115.	Protective finish inadequate or defective	Visual
116.	Surface finish of midsection internal diameter and small internal diameter, incorrect	Visual
117.	Burrs or sharp corners	Visual
118.	Foreign matter	Visual

MINOR

201.	Pitch diameter of internal thread, min.	Gage
202.	Pitch diameter of external thread, max.	Gage
203.	Inside diameter small end of body	Gage
204.	Diameter of external thread undercut	Gage
205.	Diameter of body large end	Gage
206.	Depth to bottom of midsection internal diameter, min.	Gage
207.	Length of body small end	Gage
208.	Total length	Gage
209.	Width across flats	Gage
210.	Outside diameter of body, small end	Gage
211.	Depth of inside diameter, small end of body	Gage
212.	Width of internal thread undercut	Gage
213.	Width of external thread undercut	Gage
214.	Location of external thread undercut	Gage
215.	Location of flats	Gage
216.	Radius or chamfers, incorrect	Gage
217.	Radius or chamfers missing, except as otherwise classified	Visual
218.	Surface finish incorrect, except as otherwise classified	Visual
219.	Part number missing, incorrect or illegible	Visual

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4.3.2.3 Disc, Shear, Drawing B8597501 covering a detail of Drawing F8597515.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
101. Thickness	Gage
102. Diameter of hole, max.	Gage
103. Concentricity of hole with outside diameter	Gage
104. Metal defective	Visual
105. Foreign matter	Visual
MINOR	
201. Outside diameter	Gage
202. Diameter of hole, min.	Gage

4.3.2.4 Disc, Shear (Front), Drawing B8597502 covering a detail of Drawing F8597515.

CRITICAL	
None defined.	
MAJOR	
101. Diameter of hole, max.	Gage
102. Thickness	Gage
103. Total width	Gage
104. Metal defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter of hole, min.	Gage
202. Outside diameter	Gage
203. Surface finish incorrect	Visual

4.3.2.5 Head, Drawing F8597503 covering a detail of Drawing F8597515.

CRITICAL	
None defined.	
MAJOR	
101. Pitch diameter of external thread, min.	Gage
102. Major diameter of external thread, min.	Gage
103. Pitch diameter of small internal thread, max.	Gage
104. Minor diameter of small internal thread, max.	Gage
105. Pitch diameter of large internal thread, max.	Gage
106. Minor diameter of large internal thread, max.	Gage
107. Concentricity of midsection inside diameter with long inside diameter	Gage

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4.3.2.5 (Cont) MAJOR

	Defect	Method of Inspection
108.	Concentricity of large internal thread with long inside diameter	Gage
109.	Concentricity of internal boss with long inside diameter	Gage
110.	True position of shear pin hole to long inside diameter	Gage
111.	Diameter of internal boss	Gage
112.	Long inside diameter	Gage
113.	Diameter of shear pin hole	Gage
114.	Midsection inside diameter	Gage
115.	Diameter of counterbore, small end	Gage
116.	Length of small internal thread, max.	Gage
117.	Effective length of small internal thread, min.	Gage
118.	Location of shear pin hole	Gage
119.	Effective length of large internal thread, min.	Gage
120.	Height of internal boss	Gage
121.	Small inside diameter	Gage
122.	Depth of counterbore, small end	Gage
123.	Depth to small diameter, small end	Gage
124.	Diameter of flare angle, small end	Gage
125.	Angle of chamfer, front of counterbore, small end	Gage
126.	Angle of chamfer, small internal thread	Gage
127.	Threads damaged	Visual
128.	Protective finish inadequate or defective	Visual
129.	Surface finish on counterbore and chamfers, small end, long inside diameter and midsection inside diameter, incorrect	Visual
130.	Burrs or sharp corners	Visual
131.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of external thread, max.	Gage
202.	Pitch diameter of small internal thread, min.	Gage
203.	Pitch diameter of large internal thread, min.	Gage
204.	Pitch diameter of set screw thread	Gage
205.	Minor diameter of set screw thread, max.	Gage
206.	Outside diameter body, large end	Gage
207.	Outside diameter of shoulder, large end	Gage
208.	Outside diameter of midsection	Gage
209.	Outside diameter of shoulder where shear pin hole is located	Gage
210.	Inside diameter of large thread undercut, max.	Gage

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	Defect	Method of Inspection
211.	Depth to bottom of long inside diameter	Gage
212.	Depth to bottom of internal boss	Gage
213.	Depth to midsection inside diameter, small end	Gage
214.	Depth of set screw hole, max.	Gage
215.	Length of effective thread, min, in shear pin hole	Gage
216.	Length from small outside shoulder to large outside shoulder	Gage
217.	Outside diameter of midsection	Gage
218.	Depth to small inside thread	Gage
219.	Depth of mounting cutout	Gage
220.	Total length	Gage
221.	Length to large body	Gage
222.	Length to large shoulder	Gage
223.	Length to shear pin hole shoulder	Gage
224.	Width across flats	Gage
225.	Width of mounting cutout	Gage
226.	Width of external undercut	Gage
227.	Width of internal undercut, max.	Gage
228.	Location of flat	Gage
229.	Radius or chamfers incorrect	Gage
230.	Radius or chamfers missing	Visual
231.	Surface finish incorrect, except as otherwise classified	Visual
232.	Part number missing, incorrect or illegible	Visual
233.	Part number not applied prior to protective finish	Visual

4.3.2.6 Pin, Firing, Drawing C8597504 covering a detail of Drawing F8597515.

CRITICAL

None defined.

MAJOR

101.	Concentricity of striker with large body	Gage
102.	Concentricity of small body with large body	Gage
103.	Concentricity of groove with large body	Gage
104.	Perpendicularity of striker face with large body	Gage
105.	True position of pin hole with large body	Gage
106.	Diameter of pin hole, max.	Gage
107.	Diameter of striker	Gage
108.	Diameter of body small end, max.	Gage
109.	Diameter of body large end, min.	Gage
110.	Diameter of groove	Gage
111.	Depth of pin hole, max.	Gage

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4.3.2.6 (Cont) MAJOR

	Defect	Method of Inspection
112.	Length of striker	Gage
113.	Location of pin hole	Gage
114.	Profile of groove	Gage
115.	Metal defective	Visual
116.	Surface finish of body	Visual
117.	Burrs or sharp corners	Visual
118.	Foreign matter	Visual

MINOR

201.	Diameter of body, large end, max.	Gage
202.	Diameter of body, small end, min.	Gage
203.	Diameter of pin hole, min.	Gage
204.	Depth of pin hole, min.	Gage
205.	Depth of slot in base	Gage
206.	Length of small body diameter	Gage
207.	Total length	Gage
208.	Width of groove	Gage
209.	Width of slot in base	Gage
210.	Location of groove	Gage
211.	Location of slot in base	Gage
212.	Radius incorrect	Gage
213.	Radius missing	Visual
214.	Surface finish incorrect, except as otherwise specified	Visual
215.	Part numbers missing, incorrect or illegible	Visual
216.	Part number not applied prior to protective finish	Visual

4.3.2.7 Pin, Shear, Drawing B8597505 covering a detail of Drawing F8597515.

CRITICAL

None defined.

MAJOR

101.	Diameter, max.	Gage
102.	Final diameter (after plating) max.	Gage
103.	Total length, min.	Gage
104.	Metal defective	Visual
105.	Protective finish inadequate or defective	Visual
106.	Surface finish incorrect	Visual
107.	Burr	Visual
108.	Foreign matter	Visual

MINOR

201.	Total length, max.	Gage
202.	Profile of end	Gage

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4.3.2.8 Piston, Gas, Drawing C8597506 covering a detail of Drawing F8597515.

	Defect	Method of Inspection
CRITICAL		
	None defined.	
MAJOR		
101.	Concentricity of groove with body	Gage
102.	Perpendicularity of face, threaded end, with body	Gage
103.	Diameter of body, min.	Gage
104.	Diameter of groove	Gage
105.	Total length	Gage
106.	Profile of groove	Gage
107.	Metal defective	Visual
108.	Protective finish inadequate or defective	Visual
109.	Burr or sharp corners	Visual
110.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread	Gage
202.	Minor diameter of thread, max.	Gage
203.	Diameter of body, max.	Gage
204.	Diameter of boss	Gage
205.	Depth of thread, min.	Gage
206.	Width of groove	Gage
207.	Location of groove	Gage
208.	Height of boss	Gage
209.	Radius or chamfer incorrect	Gage
210.	Radius or chamfer missing	Visual
211.	Part number missing, incorrect or illegible	Visual
212.	Part number not applied prior to protective finish	Visual

4.3.2.9 Piston End Assembly, Drawing D8597508 covering a detail of Drawing F8597515.

CRITICAL		
	None defined.	
MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of pitch diameter of thread with shoulder	Gage
104.	Concentricity of body with shoulder	Gage
105.	Concentricity of head with shoulder	Gage
106.	Diameter of shoulder	Gage
107.	Diameter of body	Gage
108.	Diameter of head	Gage

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4.3.2.9 (Cont) MAJOR

	Defect	Method of Inspection
109.	Length to body, max.	Gage
110.	Length of body, max.	Gage
111.	Protrusion of pellet above thread (after plating)	Gage
112.	Metal defective	Visual
113.	Protective finish, improper	Visual
114.	Surface finish of body, incorrect	Visual
115.	Burr	Visual
116.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, max.	Gage
202.	Diameter of thread clearance cut	Gage
203.	Length to body, min.	Gage
204.	Length of body, min.	Gage
205.	Total length	Gage
206.	Width of undercut	Gage
207.	Width across flats	Gage
208.	Location of undercut	Gage
209.	Length of flats	Gage
210.	Length to flats	Gage
211.	Location of flats	Gage
212.	Location of pellets	Gage
213.	Radius or chamfer incorrect	Gage
214.	Radius or chamfer missing	Visual
215.	Pellets missing (after plating)	Visual
216.	Profile of head	Visual
217.	Surface finish incorrect, except as otherwise classified	Visual
218.	Part number missing, incorrect or illegible	Visual

4.3.2.10 Plug, Drawing C8597509 covering a detail of Drawing F8597515.

CRITICAL		
	None defined.	
MAJOR		
101.	Diameter of body	Gage
102.	Metal defective	Visual
103.	Surface finish on body and face of shoulder, incorrect	Visual
MINOR		
201.	Diameter of undercut, min.	Gage
202.	Width of undercut	Gage
203.	Length of body	Gage

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4.3.2.10 (Cont) MINOR

	Defect	Method of Inspection
204.	Diameter of bore	Gage
205.	Diameter to shoulder	Gage
206.	Total length	Gage
207.	Radius or chamfer, incorrect	Gage
208.	Radius or chamfer, missing	Visual
209.	Surface finish, improper, except as otherwise classified	Visual

4.3.2.11 Ring, Drawing D8597510 covering a detail of Drawing F8597515.

CRITICAL
None defined.

MAJOR

101.	Concentricity of body with bore	Gage
102.	Concentricity of external groove with bore	Gage
103.	Concentricity of internal groove with bore	Gage
104.	Parallelism of faces	Gage
105.	Perpendicularity of face with bore	Gage
106.	Diameter of body	Gage
107.	Diameter of external groove	Gage
108.	Diameter of internal groove	Gage
109.	Diameter of bore	Gage
110.	Total length	Gage
111.	Profile of external groove	Gage
112.	Profile of internal groove	Gage
113.	Metal defective	Visual
114.	Protective finish inadequate or defective	Visual
115.	Surface finish, improper	Visual
116.	Burr or sharp corner	Visual
117.	Foreign matter	Visual

MINOR

201.	Width of external groove	Gage
202.	Width of internal groove	Gage
203.	Location of external groove	Gage
204.	Location of internal groove	Gage
205.	Radius or chamfer, incorrect	Gage
206.	Radius or chamfer, missing	Visual
207.	Part number missing, incorrect or illegible	Visual

4.3.2.12 Body Assembly Drawing B8597516 covering a detail of Drawing F8597515.

CRITICAL
None defined.

MAJOR

101.	Pitch diameter of small thread, min.	Gage
102.	Major diameter of small thread, min.	Gage

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4.3.2.12 (Cont) MAJOR

	Defect	Method of Inspection
103.	Pitch diameter of large thread, min.	Gage
104.	Major diameter of large thread, min.	Gage
105.	Concentricity of pitch diameter of small thread with bore	Gage
106.	Concentricity of pitch diameter of large thread with bore	Gage
107.	Concentricity of counterbore with bore	Gage
108.	Diameter of counterbore	Gage
109.	Diameter of bore	Gage
110.	Wall thickness of body	Gage
111.	Length of effective thread, large end, min.	Gage
112.	Perpendicularity of bore to face, large thread end	Gage
113.	Straightness of bore	Gage
114.	Radius at base of counterbore	Gage
115.	Protrusion of pellets above thread (after plating)	Gage
116.	Threads damaged	Visual
117.	Protective finish inadequate or defective	Visual
118.	Metal defective	Visual
119.	Burrs or sharp corner	Visual
120.	Foreign matter	Visual

MINOR

201.	Pitch diameter of small thread, max.	Gage
202.	Pitch diameter of large thread, max.	Gage
203.	Diameter of body	Gage
204.	Diameter of undercut, large thread, if applicable	Gage
205.	Diameter of thread lead, large end	Gage
206.	Length of thread lead, small end	Gage
207.	Wall thickness at thread lead, small end	Gage
208.	Diameter of pellet holes	Gage
209.	Depth of pellet holes	Gage
210.	Depth of counterbore	Gage
211.	Total length	Gage
212.	Length of thread lead, large end	Measure
213.	Width of small thread undercut	Measure
214.	Width of large thread undercut, if applicable	Measure
215.	Location of small thread undercut	Measure
216.	Location of pellet holes	Measure
217.	Radius or chamfer, incorrect	Visual
218.	Radius or chamfer, missing	Visual
219.	Pellets missing	Visual
220.	Part number missing, incorrect or illegible	Visual

4.3.2.13 Body, Piston, Drawing C8597517 covering a detail of Drawing F8597515.

CRITICAL

None defined.

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4.3.2.13 (Cont)

Defect	Method of Inspection
MAJOR	
101. Large inside diameter	Gage
102. Concentricity of large inside diameter to outside diameter	Gage
103. Total length	Gage
104. Surface finish large inside diameter and face, incorrect	Visual
105. Metal defective	Visual
MINOR	
201. Small inside diameter	Gage
202. Concentricity of small inside diameter to outside diameter	Gage
203. Length of large inside diameter	Gage
204. Depth of cavity	Gage
205. Length and angle of chamfer	Gage
206. Foreign matter	Visual

4.3.2.14 Piston Rod Assembly, Drawing F8597518 covering a detail of Drawing F8597515.

CRITICAL	
None defined.	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of pitch diameter with body	Gage
104. Concentricity of small counterbore with thread	Gage
105. Concentricity of large counterbore with thread	Gage
106. Concentricity of head with body	Gage
107. Concentricity of groove with head	Gage
108. Straightness of body	Gage
109. Perpendicularity of face to body, threaded end	Gage
110. True position of top relief hole with relief hole in face of head	Gage
111. Diameter of small counterbore	Gage
112. Diameter of large counterbore	Gage
113. Diameter of body	Gage
114. Diameter of head	Gage
115. Diameter of groove	Gage
116. Diameter of relief hole in face	Gage
117. Diameter of top relief hole	Gage
118. Depth of relief hole in face, min.	Gage
119. Depth of top relief hole, min.	Gage
120. Depth of large counterbore, min.	Gage
121. Depth to bottom of small counterbore, min.	Gage
122. Total length, min.	Gage
123. Profile of groove	Gage

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4.3.2.14 (Cont) MAJOR

	Defect	Method of Inspection
124.	Metal defective	Visual
125.	Protective finish inadequate or defective	Visual
126.	Surface finish, improper	Visual
127.	Burr or sharp edge	Visual
128.	Foreign matter	Visual

MINOR

201.	Pitch diameter of thread, min.	Gage
202.	Diameter of head undercut	Gage
203.	Diameter of wrench holes in face	Gage
204.	Depth of relief hole in face, max.	Gage
205.	Depth of top relief hole, max.	Gage
206.	Depth of large counterbore, max.	Gage
207.	Depth to bottom of small counterbore, max.	Gage
208.	Total length, max.	Gage
209.	Length of flange	Gage
210.	Location of groove	Gage
211.	Depth of wrench holes in face	Gage
212.	Width of head undercut	Gage
213.	Location of wrench holes in face	Gage
214.	Location of relief hole in face	Gage
215.	Location of top relief hole	Gage
216.	Location of head undercut	Gage
217.	Radius or chamfers incorrect	Gage
218.	Radius or chamfers missing	Visual
219.	Part number, missing, incorrect or illegible	Visual

4.3.2.15 Tube, Drawing D8597519 covering a detail of Drawing F8597515.

CRITICAL

None defined.

MAJOR

101.	Concentricity of small inside diameter to body	Gage
102.	Perpendicularity of lip flanged end to body	Gage
103.	Flatness of flange	Gage
104.	Straightness of body	Gage
105.	Diameter, inside	Gage
106.	Diameter, outside	Gage
107.	Diameter of flange	Gage
108.	Thickness of flange, min.	Gage
109.	Metal defective	Visual
110.	Protective finish inadequate or defective	Visual
111.	Radius or chamfer incorrect	Gage
112.	Radius or chamfer missing	Visual
113.	Burr or sharp edges	Visual
114.	Foreign matter	Visual

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4.3.2.15 (Cont)

Defect	Method of Inspection
MINOR	
201. Diameter of hole	Gage
202. Total length	Gage
203. Surface finish, incorrect	Visual

4.3.2.16 Tube Assembly, Drawing D8597520 covering a detail of Drawing F8597515.

CRITICAL
None defined.

MAJOR	
101. Concentricity of bore of shear disc to bore of tube	Gage
102. Solder buildup on the inside	Visual
103. Solder drippings on outside	Visual
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Foreign matter	Visual

MINOR
None defined.

4.3.2.17 Assembly, Drawing F8597515.

CRITICAL	
1. Evidence of oil leakage	Visual
MAJOR	
101. Thread damaged	Visual
102. Lot number missing, incorrect or illegible	Visual
103. Metal defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Shipping cap missing	Visual
202. Protective finish inadequate or defective	Visual

4.3.2.18 Packing in Fiber Container, Drawing C8848865.

CRITICAL
None defined.

MAJOR
None defined.

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4.3.2.18 (Cont)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damaged through all impregnated layers	Visual

4.3.2.19 Packing in Wooden Box, Drawing C8848864.

CRITICAL
None defined.

MAJOR
None defined.

MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual and Manual
203. Strapping missing, loose, broken or improperly assembled	Visual and Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

4.3.2.20 Process control.- Periodic unscheduled monitoring shall be conducted for the following process operations to determine compliance with the applicable requirements of the indicated drawing:

Amount of silicone fluid - - F8597515
Bottoming torque - - Requirement 3.7.2

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4:

4.3.3.1 Hardness.- Fifty units from each heat treated lot of components shall be tested to determine compliance with the requirements noted on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.3.2 Hydrostatic pressure.- Each body, head and cap shall be subjected to an internal hydrostatic pressure test. Failure of any unit to comply with the requirements of 3.5 shall be cause for rejection of the component.

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4.3.3.3 Shear pin strength.- Fifty pins from each lot shall be tested for shear strength. Failure of any pin to comply with the requirement of 3.6 shall be cause for rejection of the lot.

4.3.3.4 Breakaway torque.- Each body and cap subassembly shall be subjected to a breakaway torque as listed in Table I. Failure of any subassembly to comply with the requirements of 3.7.2 shall be cause for rejection of the subassembly.

4.3.3.5 Protective finish.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standards listed on the drawing.

4.3.3.6 Residual magnetism.- Each metal parts' assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirement of 3.9 shall be cause for rejection of the assembly.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8597515.

4.4 Test methods and procedures.

4.4.1 Hardness.- Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the hardness requirements specified on the detail drawing.

4.4.2 Hydrostatic pressure.- Prior to the application of protective finish, the body, cap and head, individually, shall be mounted in a device and subjected to an internal hydrostatic pressure in accordance with the requirements of 3.5. The testing device used shall internally seal the body, cap and head, as applicable, at each end without applying any axial load other than that applied by the liquid pressure.

4.4.3 Shear pin strength.- After application of protective finish, the pins shall be subjected to the shear pin strength test in accordance with the requirements of 3.6. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.4.4 Breakaway torque.- A torque wrench, set at the torque requirement specified in 3.7.2, shall be used in conjunction with a fixture to determine the adequacy of the joint. The body shall be secured in an air vise or other fixture and the breakaway torque shall be applied to the cap.

4.4.5 Protective finish.- Test procedures for protective finish shall be in accordance with the applicable documents referenced on the component drawing.

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4.4.6 Residual magnetism.- The metal parts' assembly shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the assembly 6 inches from and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading as shown in figure 1. The assembly shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90-degree rotation of the assembly.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with Drawings C8848864 and C8848865.

5.2 Data cards.- Data card information shall be as specified in MIL-A-2550.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:

Army - MU

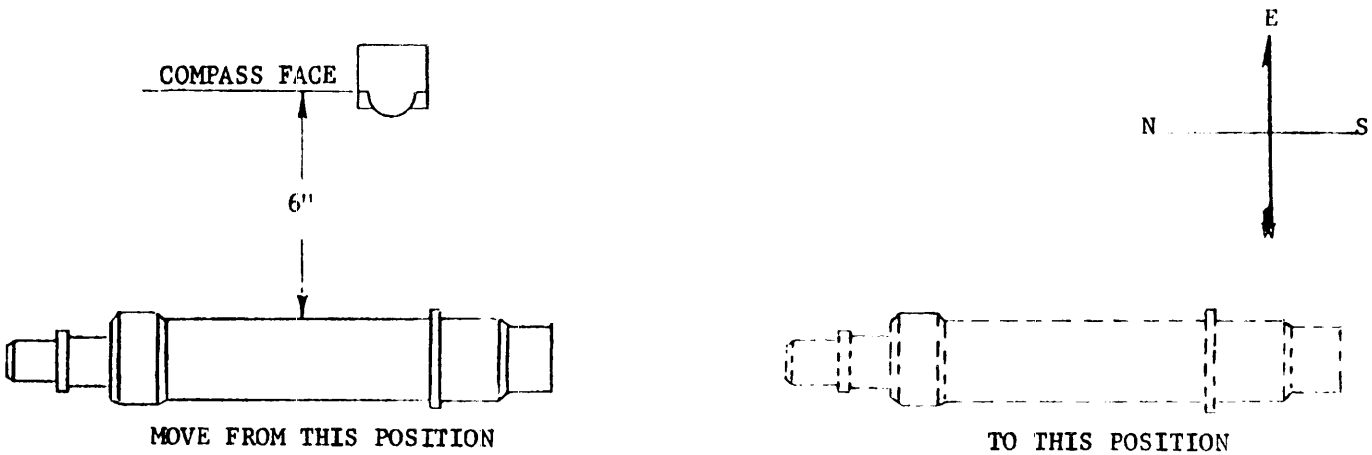
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PILOT'S STANDBY COMPASS, TYPE AN-5766



PLAN VIEW

NOTE:

THRUSTER AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

FIGURE 1

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN THRUSTER, CARTRIDGE ACTUATED, M18

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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